

Declaration

We hereby declare that the Capstone Project entitled “**G-Hydro HLM: A web based Hostel Leave Management System for GHEC**” submitted by us for the Degree of Bachelor of Engineering in Computer Science and Engineering is the result of our original and independent research work carried out under the guidance of Supervisor Dr. Shashi Gurung and it has not been submitted to any other university or institute for the award of any degree or diploma. We also hereby declare that this written submission represents my own ideas in my own words. We also declare that the other ideas, which are included in project, have adequately provided the reference from the original reported work.

We have taken care in all respect to honor the intellectual Property Right and have acknowledged the contribution of others for using them in this project work for academic purpose. We further declare that in case of any violation of intellectual Property Right or Copy Right, We as the candidate will be fully responsible for the same. My Supervisor, Head of the Department and Institute Head should not be held responsible for any violation of copy right if found at any stage of my degree.

Place Bandla

Date May, 2026

Signature of the Students

Certificate

This is to certify that the Capstone Project entitled “**G-Hydro HLM: A web based Hostel Leave Management System for GHEC**” submitted by **Priyal Sood**(23010503043), **Ritisha Dass**(23010503049), **Sakshi**(23010503050), **Sourav Kumar**(23010503065), **Varnika Samkaria**(23010503073) for the award of the degree of the Bachelor of Engineering in Computer Science and Engineering, Government Hydro Engineering College Bandla, Bilaspur, Himachal Pradesh, India is entirely based on the work carried out by him under our supervision and guidance, The work recorded, embodies the original work of the candidate and has not been submitted for the award of any degree, or diploma of any university and institute, according to the best of our knowledge.

Date May, 2026

Place GHEC Bandla

Signature of Supervisor

Acknowledgement

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ABSTRACT

The **G-Hydro HLM (Hostel Leave Management System)** is a web-based application developed to modernize and simplify hostel leave management at Government Hydro Engineering College (GHEC). Traditional hostel leave systems rely heavily on paper-based processes, which are often time-consuming, inefficient, and prone to errors such as record loss, unauthorized modifications, and delays in approval. These limitations create difficulties for both students and hostel authorities in managing leave requests and monitoring student movement effectively.

The primary objective of the G-Hydro HLM system is to digitize the hostel leave process by providing a centralized platform where students can apply for leave online, and hostel authorities can review, approve, or reject requests efficiently. The system incorporates a structured multi-level approval mechanism involving the Superintendent and Warden to ensure proper authorization and transparency in hostel administration.

The proposed system has been developed using modern web technologies, including **HTML5, CSS3, Bootstrap, Python, Flask, and SQLite**, to provide a secure and responsive user experience. The implementation of **Role-Based Access Control (RBAC)** ensures that each user has access only to functionalities relevant to their role, such as student, superintendent, warden, or director. Security measures such as **Bcrypt password encryption** and **SSL/TLS communication protocols** have been integrated to protect sensitive student data and maintain system integrity.

A key feature of the system is the integration of a **QR-code-based verification mechanism** that enables real-time hostel gate monitoring and movement tracking. Students receive unique digital QR passes after leave approval, allowing hostel authorities to quickly record entry and exit activities. Additionally, a **Manual UID Override mechanism** has been incorporated to ensure uninterrupted operation during technical failures or emergency situations.

The system was tested under multiple scenarios, including functional testing, integration testing, security testing, and stress testing, and demonstrated reliable performance with reduced leave processing time and improved administrative efficiency. By replacing traditional manual methods with a secure digital framework, G-Hydro HLM significantly improves hostel management, enhances transparency, strengthens security, and provides a scalable foundation for future smart hostel solutions.

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GLOSSARY

Term / Abbreviation	Definition
ACID	Atomicity, Consistency, Isolation, Durability — A set of properties that guarantee database transactions are processed reliably.
API	Application Programming Interface — A set of rules that allows different software components to communicate with each other.
Bcrypt	A cryptographic password hashing function based on the Blowfish cipher, used to securely store user credentials.
Bootstrap	An open-source frontend framework used for designing responsive and mobile-friendly web interfaces.
CSS3	Cascading Style Sheets (version 3) — A stylesheet language used for describing the visual appearance of web pages.
DFD	Data Flow Diagram — A graphical representation of the flow of data through an information system.
ERD	Entity-Relationship Diagram — A structural diagram used to design and represent relationships between database tables.
Flask	A lightweight Python-based web framework used for building backend server logic and routing.
GHEC	Government Hydro Engineering College, Bandla, Bilaspur, Himachal Pradesh.
G-Hydro HLM	G-Hydro Hostel Leave Management System — The proposed web-based digital system developed for GHEC.
HTML5	HyperText Markup Language (version 5) — The standard language used to structure web page content.

Term / Abbreviation	Definition
HTTP	HyperText Transfer Protocol — The foundation of data communication on the World Wide Web.
HTTPS	HTTP Secure — An encrypted version of HTTP that uses SSL/TLS to protect data in transit.
LAN	Local Area Network — A network connecting computers within a limited area such as a campus.
MITM	Man-in-the-Middle Attack — A cyberattack where an unauthorized party secretly intercepts communication between two parties.
RBAC	Role-Based Access Control — A security model that restricts system access based on roles assigned to individual users.
RDBMS	Relational Database Management System — A type of database that stores data in structured tables with defined relationships.
SQL	Structured Query Language — A programming language used for managing and querying relational databases.
SQLite	A lightweight, serverless relational database engine used for local data storage.
SQLAlchemy	A Python library providing an Object Relational Mapper (ORM) for interacting with databases using Python code.
SSL/TLS	Secure Sockets Layer / Transport Layer Security — Cryptographic protocols used to secure data transmission over a network.
UID	Unique Identification Number — A persistent identifier assigned to each student for backup manual verification.
UI	User Interface — The visual layout and elements through which a user interacts with a software system.
URL	Uniform Resource Locator — A web address used to access resources on the internet.
WAL	Write-Ahead Logging — A database journaling mode that allows concurrent reads while safely serializing write operations.

Term / Abbreviation	Definition
QR Code	Quick Response Code — A two-dimensional barcode that encodes data and can be scanned using a smartphone camera for fast digital verification.